

News Category Classification

A Full ML Pipeline on the AG News Dataset

Logistic Regression · Linear SVM · Random Forest · MLP Neural Network

TF-IDF Feature Engineering · 120,000 Training Articles · 4 Categories

World

Sports

Business

Sci/Tech

Outline

- 1 Introduction & Dataset
- 2 Text Preprocessing
- 3 Feature Engineering: TF-IDF
- 4 Models & Methodology
- 5 Exploratory Analysis
- 6 Results & Evaluation
- 7 Summary & Conclusions

Quick Facts

- **120K** training articles
- **7,600** test articles
- **4** balanced categories
- TF-IDF + 4 classifiers
- Best accuracy **91.82%**

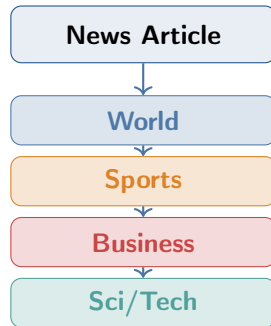
Task Overview

Problem Statement

Automatically assign a **news article** to one of four topical categories using classical ML and a shallow neural network.

Pipeline Steps

- 1 Load & explore AG News dataset
- 2 Text preprocessing (NLP pipeline)
- 3 TF-IDF feature extraction
- 4 Train 3 classical classifiers
- 5 *Bonus*: MLP Neural Network
- 6 Evaluation & visualisations
- 7 *Bonus*: Word clouds per category



AG News Dataset

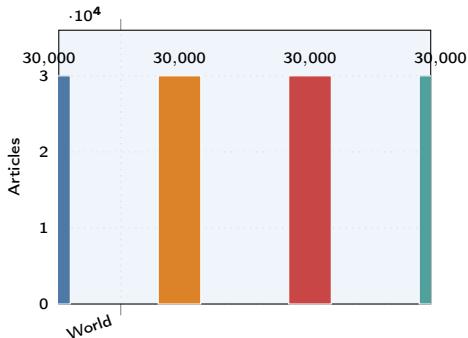
Dataset at a Glance

- Large-scale news topic classification benchmark
- Each sample: *title* + *description* combined
- 4 perfectly balanced categories (no class imbalance)

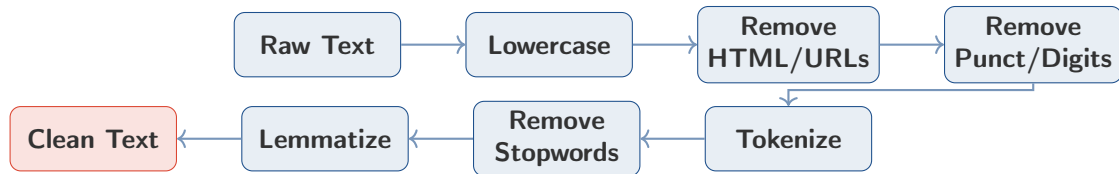
Split Summary

Split	Articles	Per class
Train	120,000	30,000
Test	7,600	1,900

Perfectly Balanced Classes



Text Preprocessing Pipeline



Steps Explained

- **Lowercase:** Normalise case
- **Remove HTML/URLs:** Strip tags and links
- **Remove Punct/Digits:** Keep only alpha chars
- **Tokenize:** Split on whitespace
- **Stopwords:** sklearn's English list
- **Lemmatize:** Custom suffix-stripping rules (no NLTK)

Before & After Example

Before:

"Iraq's PM said the attack on Baghdad was..."

After:

"iraq prime minist attack baghdad"

Why Lemmatize?

Reduces vocabulary: *running, runs, ran* → *run*. Less noise, better generalisation.

TF-IDF Feature Engineering

TF-IDF Formula

$$\text{tfidf}(t, d) = \underbrace{\log(1 + \text{tf}(t, d))}_{\text{Term Frequency}} \times \underbrace{\log\left(\frac{N}{1 + \text{df}(t)}\right)}_{\text{Inverse Doc Freq}}$$

- **TF**: How often t appears in doc d
- **IDF**: Penalises terms common across docs
- **sublinear_tf**: log-dampens high-freq terms

Vectoriser Settings

<code>max_features</code>	50,000
<code>ngram_range</code>	(1, 2) — unigrams + bigrams
<code>sublinear_tf</code>	True
<code>min_df</code>	2 (drop rare terms)

Why TF-IDF for News?

- Produces **sparse** high-dimensional vectors — ideal for linear models
- **Bigrams** capture key phrases: *“interest rate”*, *“nuclear weapon”*
- Down-weights noise words like *“said”* appearing across all categories

Matrix Properties

Train shape	120,000 × 50,000
Test shape	7,600 × 50,000
Sparsity	≈ 99.8%

No Data Leakage!

TF-IDF fitted **only on train set**, then applied to test — strictly correct ML practice.

Classifier Models

Logistic Regression

Concept: Models $P(y|x)$ via softmax.

Prediction:

$$p_k = \frac{e^{w_k^T x}}{\sum_j e^{w_j^T x}}$$

Settings:

$C = 5$, solver=SAGA

Strengths:

Fast, interpretable, strong baseline for sparse TF-IDF. L2 regularisation prevents overfitting.

Linear SVM

Concept: Max-margin hyperplane.

Objective:

$$\min_w \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$$

Settings:

$C = 1.0$, LinearSVC

Strengths:

Gold standard for text classification — excellent on sparse high-dimensional data. OvA strategy for multiclass.

Random Forest

Concept: Bagged ensemble of decision trees.

Prediction:

Majority vote of T trees using random feature subsets

Settings:

300 trees, $n_jobs=-1$

Weakness:

Struggles on very sparse high-dim data like TF-IDF

MLP Neural Net

Concept: Feedforward network.

Architecture:

$50K \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 4$

ReLU activations

Settings:

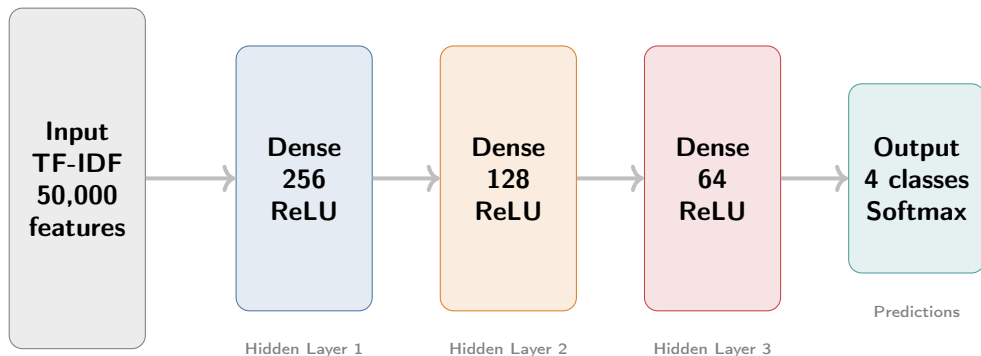
Adam, L2 ($\alpha=0.0001$)

50 epochs, early stopping (patience=5)

Strengths:

Learns complex feature interactions

MLP Neural Network — Architecture



Training Config

Optimiser: Adam
L2 reg: $\alpha = 0.0001$
Max epochs: 50
Early stopping: patience=5

Why ReLU?

$\text{ReLU}(x) = \max(0, x)$
Avoids vanishing gradient.
Sparse activations are efficient on high-dim TF-IDF input.

Early Stopping

Monitors 10% validation split. Halts if no improvement for 5 consecutive epochs — prevents overfitting.

Top Keywords per Category

World — Top Terms

said, iraq, reuter, afp, kill, president, minist,
new, people, govern

Dominated by geopolitical and conflict vocabulary

Sports — Top Terms

game, win, team, season, night, play, victory,
year, cup, world

Highly distinctive competition vocabulary

Business — Top Terms

said, reuter, new, oil, company, stock, prices,
percent, year, york

Financial and market-driven vocabulary

Sci/Tech — Top Terms

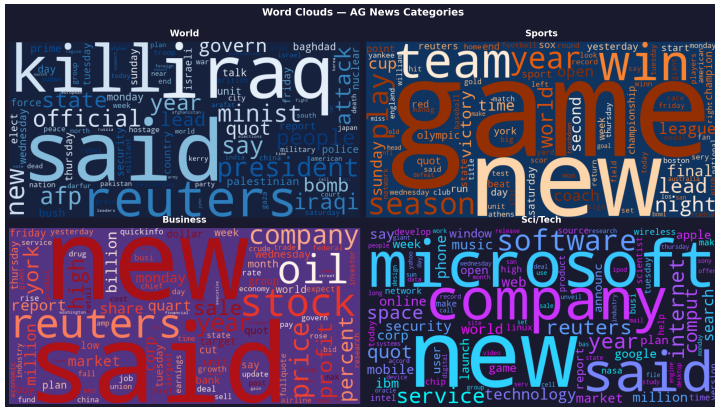
new, microsoft, company, software, internet,
space, quot, comput, search

Technology-company and innovation vocabulary

Key Observation

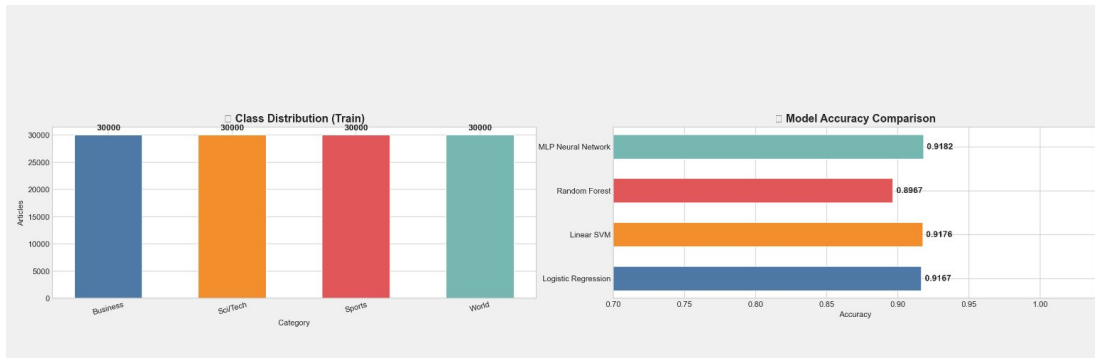
said and **reuter** appear across all categories —
discriminating features must be content-specific
terms.

Word Clouds per Category



Visually confirms distinct vocabulary per category — strong separability supports high classification accuracy. Sports and Sci/Tech are the most visually distinctive.

Results — Class Distribution & Model Accuracy



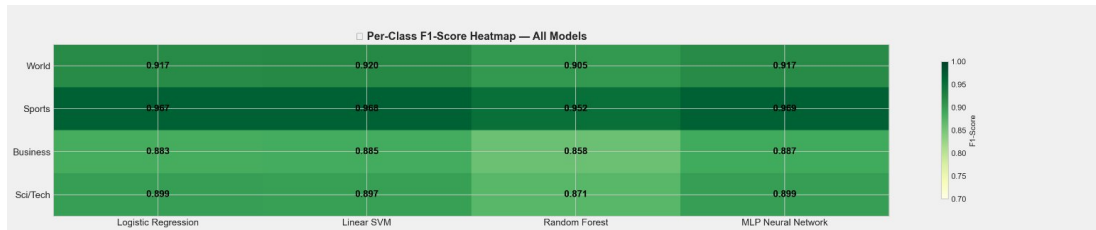
Perfectly balanced classes (30 000 each). MLP leads at 91.8% accuracy, closely followed by Linear SVM (91.8%) and Logistic Regression (91.7%). Random Forest trails at 89.7%.

Results — Top-10 Words & Confusion Matrices



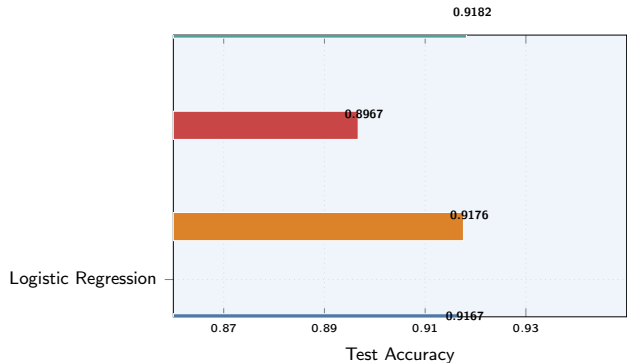
Each category has a distinct vocabulary signature. Business/World confusion is the most frequent misclassification across all models; Sports is consistently the cleanest class.

Results — Per-Class F1-Score Heatmap



Sports achieves the highest F1 (≥ 0.95) across all models. Business is the hardest class (F1 0.858–0.887). MLP and Linear SVM are the most consistent top performers.

Model Accuracy Comparison



Key Observations

- **MLP** tops at **91.82%** — learns feature interactions
- **Linear SVM** close second (91.76%) with faster training
- **Logistic Regression** competitive at 91.67%
- **Random Forest** weakest (89.67%) — tree splits are inefficient on sparse data

Winner

MLP Neural Network
91.82% test accuracy

Per-Class F1-Score Heatmap

F1-Score Heatmap — All Models

World	0.917	0.920	0.905	0.917
Sports	0.987	0.988	0.952	0.988
Business	0.883	0.885	0.858	0.887
Sci/Tech	0.899	0.897	0.871	0.899

Insights

- **Sports** F1 ≈ 0.987 — very distinctive vocabulary
- **Business** hardest (≈ 0.85 – 0.89) — overlaps with World and Sci/Tech
- **Linear models** consistently outperform Random Forest
- **MLP** matches or beats all classicals

Why is Business Hardest?

Business articles mention tech companies ($\rightarrow$ Sci/Tech confusion) and world events like oil/geopolitics ($\rightarrow$ World confusion)

Confusion Matrix — MLP Neural Network

MLP Neural Network — Test Set (7,600 articles)

		Predicted Label				True Label
		World	Sports	Business	Sci/Tech	
World	1716	53	77	34		
Sports	24	1863	9	4		
Business	66	15	1699	120		
Sci/Tech	60	15	145	1680		

Error Analysis

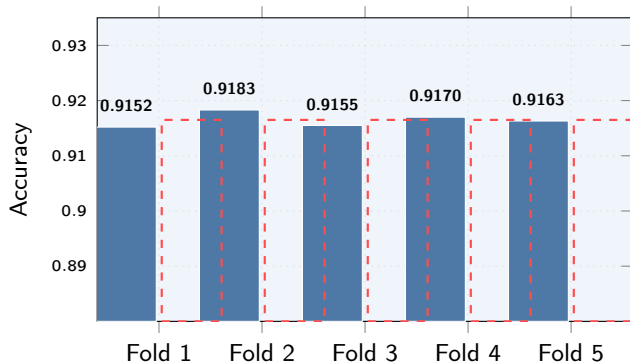
- **Sports** cleanest: only 37 misclassifications from 1,900
- **Business**→**Sci/Tech**: 120 errors — tech companies in business news
- **Sci/Tech**→**Business**: 145 errors — financial tech articles
- **World** modest confusion with Business/Sci/Tech

Per-Class Accuracy

World	90.3%
Sports	98.0%
Business	89.4%
Sci/Tech	88.4%

5-Fold Cross-Validation

CV Results — Logistic Regression



Mean: 0.9165 ± 0.0011 (5-fold, 24K subsample)

What Cross-Validation Tells Us

- Very **low variance** (± 0.0011) — model is stable
- Results are **not a lucky split**
- Mean CV $\approx$ test accuracy: **no overfitting**

K-Fold CV Method

Data split into $k=5$ equal folds. Each fold acts as validation once; model trained on remaining 4. Final score = mean of 5 validation accuracies.

Stability Confirmed

Standard deviation of 0.0011 indicates the pipeline generalises well regardless of the random data split.

Results Summary

Final Model Comparison — AG News Test Set (7,600 articles)

Model	Accuracy	World F1	Sports F1	Business F1	Sci/Tech F1
Logistic Regression	0.9167	0.917	0.987	0.883	0.899
Linear SVM	0.9176	0.920	0.988	0.885	0.897
Random Forest	0.8967	0.905	0.952	0.858	0.871
MLP Neural Network	0.9182	0.917	0.988	0.887	0.899

Best Overall

MLP Neural Network
91.82% accuracy

Best Efficiency

Linear SVM
91.76% — fastest
training

Weakest

Random Forest
89.67% — sparse data
issues

Key Takeaways & Conclusions

What Worked Well

- 1 **TF-IDF + Linear models** are extremely effective — simple, fast, accurate
- 2 **Bigrams** captured key phrases: *“interest rate”, “oil price”*
- 3 **Balanced dataset** (30K/class) eliminated class bias
- 4 **MLP** outperforms all classicals with simple 3-layer architecture
- 5 **Cross-validation** confirmed robustness — results are not chance

Lessons Learned

- Business/Sci/Tech boundary is inherently ambiguous
- Sports is most distinguishable — unique lexicon
- Random Forests struggle on sparse TF-IDF vectors
- Linear models remain very competitive with neural nets on text

Future Directions

- **BERT/DistilBERT** fine-tuning (>95% expected)
- Hyperparameter tuning (GridSearchCV)
- Character-level n-grams for robustness
- Attention-based feature importance analysis

Thank You

News Category Classification on AG News

MLP: **91.82%** · Linear SVM: **91.76%** · Logistic Regression: **91.67%** · Random Forest: **89.67%**

TF-IDF (50K features, unigrams + bigrams) · 120,000 training articles · 4 balanced categories

World

Sports

Business

Sci/Tech